

Unit 3

MC - Questions Answered
on D2L

MC	Written	Total
14 / 14	21 / 20	35 / 35

Knowledge

1) $-3 \leq \frac{6}{x}$
 $-\frac{3x}{x} \leq \frac{6}{x}$
 $0 \leq \frac{3x+6}{x}$

$0 \leq \frac{3(x+2)}{x}$

When $x \in (-\infty, -2] \cup (0, \infty)$

	$(-\infty, -2)$	$(-2, 0)$	$(0, \infty)$
$2x+2$	-	+	+
x	-	-	+
$f(x)$	+	0	+

2)

When $x \in (5, 7)$, the slope is	$\uparrow$ or $\downarrow$ (0.5)
Horizontal Asymptote(s):	$y = 8$ (0.5)
End Behaviour	$x \rightarrow \infty, f(x) \rightarrow 8^-$ $x \rightarrow -\infty, f(x) \rightarrow 8^+$ (1)
End Behaviour near vertical asymptotes	$x \rightarrow -5^+, f(x) \rightarrow -\infty$ $x \rightarrow -5^-, f(x) \rightarrow \infty$ $x \rightarrow 7^+, f(x) \rightarrow \infty$ $x \rightarrow 7^-, f(x) \rightarrow -\infty$ (1)

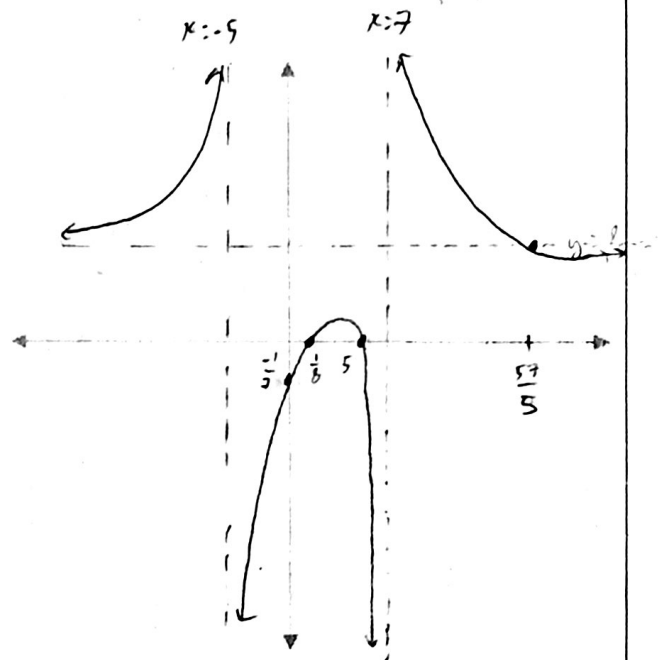
$$g(x) = \frac{8x^2 - 41x + 5}{x^2 - 2x - 35} = \frac{(2x-1)(x-5)}{(x-7)(x+5)}$$

$$\frac{8x^2 - 41x + 5}{x^2 - 2x - 35} = 8$$

$$8x^2 - 41x + 5 = 8x^2 - 16x - 280$$

$$25x - 285 = 0$$

$$x = \frac{59}{5}$$



Application

1)

$$\frac{(2x-1)(x+3)}{(x-1)(2-5)} \geq 0$$

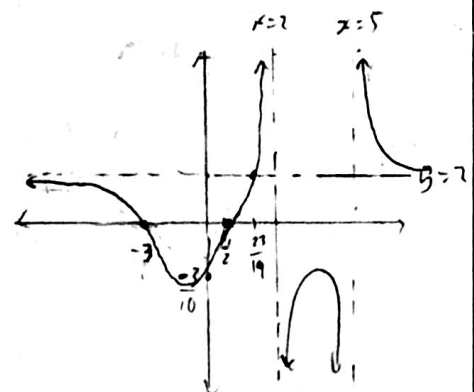
$$\frac{2x^2 + 5x - 3}{x^2 - 7x + 10} = 2$$

$$2x^2 + 5x - 3 = 2x^2 - 14x + 20$$

$$19x = 23$$

$$x = \frac{23}{19}$$

	$(-\infty, -3)$	$(-3, \frac{1}{2})$	$(\frac{1}{2}, 2)$	$(2, 5)$	$(5, \infty)$
$x+3$	-	+	+	+	+
$x-2$	-	-	-	+	+
$x-5$	-	-	-	-	+
$2x-1$	-	-	+	+	+
$f(x)$	+	0	-	+	+

Concluding statement when $x \in (-\infty, -3] \cup [\frac{1}{2}, 2) \cup (5, \infty)$

2)

(3)

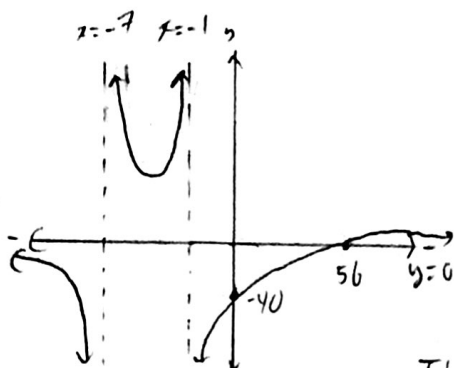
$$g(t) > f(t)$$

$$\frac{45t}{t^2 + 8t + 7} > \frac{40}{t+1}$$

$$\frac{45t}{(t+1)(t+7)} > \frac{40t+280}{(t+1)(t+7)}$$

$$\frac{5t-280}{(t+1)(t+7)} > 0$$

$$\frac{5(t-56)}{(t+1)(t+7)} > 0$$



	$(-\infty, -7)$	$(-7, -1)$	$(-1, 56)$	$(56, \infty)$
$t+7$	-	+	+	+
$t+1$	-	-	+	+
$t-56$	-	-	-	+
	-	+	-	+

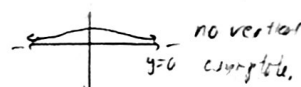
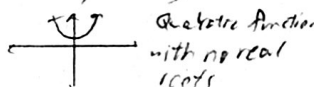
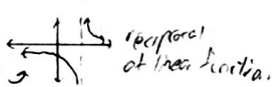
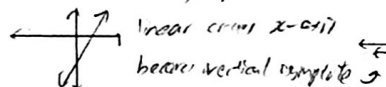
There will be more frogs than tadpoles when $t > 56$.

Communication

1)

(2)

Graphs of the reciprocal of linear functions always has a vertical asymptote (assuming it is not a horizontal line of $y=a$) because a linear function always has a 0, which cannot be in a denominator. On the other hand, quadratic functions can have no real solutions, leaving x to be any number in the denominator.



2)

(1)

Mr. Azar is not correct as if you factor the function, $\frac{(x+7)(x-2)}{x+7}$, one of the factors cancel out, leaving a linear function with a hole at $x = -7$.

Thinking

1)

$$f(x) = \frac{x^2 + x^2 - 5x + 3}{x^2 + 5x + 6}$$

$$\begin{array}{r} 1 \ 1 \ -5 \ 3 \\ \downarrow \ 1 \ 2 \ -3 \\ 1 \ 2 \ -3 \ 0 \end{array}$$

$$VA: x = -2$$

$$HA: none$$

$$OA: y = x - 4$$

$$\text{hole: } x = -3, y = -16$$

$$x\text{-int: } (1, 0)$$

$$y\text{-int: } (0, \frac{1}{2})$$

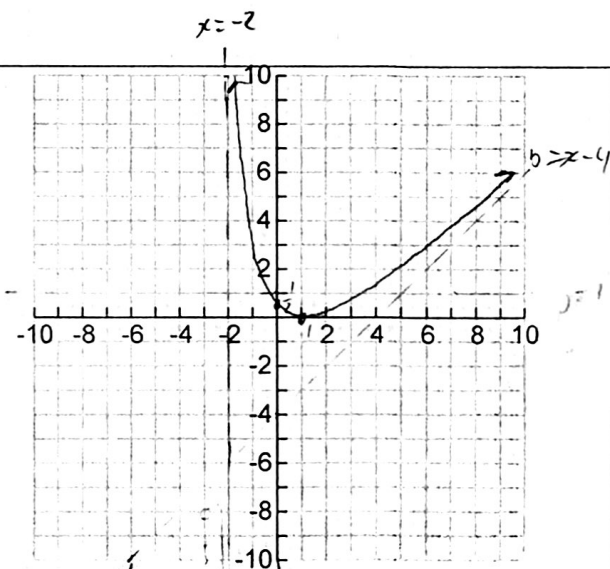
(4)

$$= \frac{(x-1)(x+3)(x-1)}{(x+2)(x+7)}$$

$$= \frac{(x-1)^2}{(x+2)}$$

$$= \frac{x^2 - 2x + 1}{x+2}$$

$$\begin{array}{r} x-4 \\ x+2 \overline{) x^2 - 2x + 1} \\ \underline{-1x^2 + 2x} \\ -4x + 1 \\ \underline{-1x^2 + 2x - 8} \\ 9 \end{array}$$



2)

(1)

$$\frac{ax^2 + b}{x^2 - 5x + 4}$$

$$\frac{a}{1} = 2$$

$$a = 2$$

$$\frac{f(0)}{4} = \frac{b}{4}$$

$$b = 1$$

$$a + b = 3$$

